PART 7 | PREDICTIVE ANALYTICS, FORECASTING AND CLUSTERING

Predictive Analytics Report

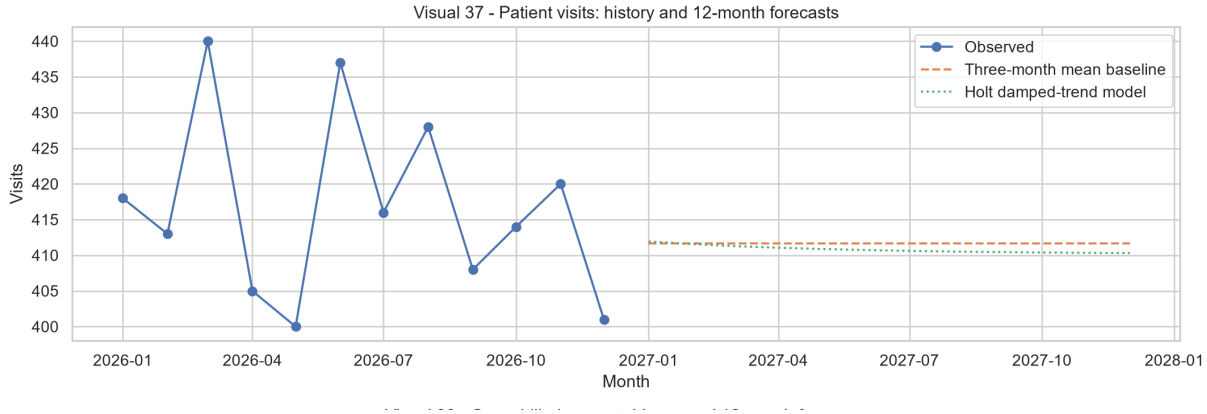
Baselines and models tested against held-out data, with deployment limits

Forecast design

The dataset contains only 12 monthly observations. Annual seasonality cannot be estimated reliably from one annual cycle, so two transparent methods are compared: a trailing three-month mean baseline and Holt's damped-trend exponential smoothing model. A rolling one-step backtest evaluates the final three observed months. Forecasts are scenario baselines, not validated annual plans.

Series	3-month mean MAE / RMSE	Holt damped trend MAE / RMSE	Lower-error method
Visits	6.56 / 7.98	7.29 / 9.69	3-month mean
Gross billed amount	900,199 / 967,054	1,600,514 / 1,824,505	3-month mean
Recorded receipts	1,542,975 / 1,830,604	1,819,092 / 2,223,477	3-month mean

The recent-mean baseline projects approximately 412 visits, 62.81 million gross billed, and 56.84 million recorded receipts per month. These are in unspecified source units. The backtest favors the simpler mean on all three series; planning should include broad uncertainty ranges and a refreshed monthly model as more history arrives.



Visual 37 patient visits

Classification results

Task	Holdout result	Assessment
Admission prediction	N=1,111; prevalence 11.5%; accuracy 0.515; balanced accuracy 0.468; ROC AUC 0.466; positive recall 0.406; precision 0.101.	Weak discrimination and many false positives (463). Not deployable for admission decisions.
Satisfaction prediction	N=419; 5 rating classes; accuracy 0.203; balanced accuracy 0.201.	Near chance for a five-class task. Class F1 values are low and similar.

The models use a holdout split and preprocessing pipelines, but performance does not support operational use. Admission is an imbalanced outcome and the classifier barely improves on weak baselines. Satisfaction ratings are noisy and only 35.1% of appointments have feedback, introducing selection bias. Feature availability at prediction time and external validation are also required before any prospective use.

Clustering and dimensionality reduction

Population	Selected result	Operational reading
Patients	K-means k=3; silhouette 0.411	Clusters summarize recorded utilization and chronic-history patterns; they are not clinical risk classes.
Facilities	Three groups of 8, 5, and 7 facilities	Mean satisfaction ranges 2.86-2.95; admission rate 10.5%-13.1%; small differences and mapping caveats remain.
Departments	Three groups of 100, 50, and 50	Average volume and satisfaction are close; roster proxies do not measure staffed capacity.
PCA	First two components explain 50.7% variance	Useful for a two-dimensional view, while half the variance remains outside the projection.

Patient cluster profiles suggest differences in documented chronic history and prior appointment counts. One group has substantial unknown chronic history, and the engineered high-risk proxy is based on age/recorded indicators. These clusters can guide data-quality review and descriptive follow-up only. They should not be used to label or triage individual patients.

Next steps

• Extend monthly history and record a complete, consistent billing and payment ledger.

• Establish temporal validation and compare against simple baselines before retraining.

• Collect clinically validated outcomes and evaluate fairness, calibration, and subgroup errors.

• Treat clusters as exploratory segments and validate with domain experts and operational evidence.